Started on	2023-11-06 02:28
State	Finished
Completed on	2023-11-06 06:37
Time taken	4 hours 9 mins
Grade	**3.00** out of 5.00 (**60**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In the solution of the following initial value problem:

$$y'' - y' - 2y = 0, \quad y(0) = -2, \quad y'(0) = 5.$$

indicate the steps with mistake(s). Check each step independently of mistakes in other steps.

Solution:

Step 1: Applying the LT to both sides of the equation we obtain

$$\mathscr{L}\{y'' - y' - 2y\}(s) = \mathscr{L}\{0\}(s)$$

Now using linearity and that $\mathscr{L}\{0\}(s) = 0$ of LT we have

$$\mathscr{L}\{y''\}(s) - \mathscr{L}\{y'\}(s) - 2\mathscr{L}\{y\}(s) = 0.$$

Step 2: Applying the LT of derivative formula and denoting for simplicity $Y(s) = \mathscr{L}\{y\}(s)$, we arrive at

$$\mathscr{L}\{y''\}(s) = s^2Y(s) - 5s + 2$$

and

$$\mathscr{L}\{y'\}(s) = sY(s) + 2.$$

Step 3: Substituting the latter in the equation and rearranging we find that

$$s^2Y(s) - 5s + 2 - sY(s) - 2 - 2Y(s) = 0$$

or

$$(s^2 - s - 2)Y(s) = 5s$$

or

$$Y(s) = \frac{5s}{(s-2)(s+1)}.$$

Step 4: The method of partial fractions for nonrepeated linear factors yields

$$Y(s) = \frac{5}{3}\left(\frac{2}{s-2} + \frac{1}{s+1}\right)$$

Step 5: Finally, inverse LT gives

$$y(t) = \mathscr{L}^{-1}\{Y(s)\}(t) = \mathscr{L}^{-1}\left\{\frac{5}{3}\left(\frac{2}{s-2} + \frac{1}{s+1}\right)\right\}(t) =$$

$$= \frac{10}{3}\mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\}(t) + \frac{5}{3}\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\}(t) = \frac{10}{3}e^{-2t} + \frac{5}{3}e^{-t}$$

Among the answers below choose the steps with mistake(s).
Solve this problem using the approach of LT and get the correct answer: $y(t) = e^{2t} - 3e^{-t}$.

Select one or more:

- [ ] Step 1
- [x] Step 2 ✓
- [ ] Step 3
- [ ] Step 4
- [x] Step 5 ✓
- [ ] there are no mistakes

Your answer is correct.

The correct answers are: Step 2, Step 5

Question 2

Incorrect

Mark 0.00 out of 1.00

In the solution of the following initial value problem:

$$ty'' - ty' + y = 2, \quad y(0) = 2, \quad y'(0) = -1.$$

indicate the steps with mistake(s). Check each step independently of mistakes in other steps.

Solution:

Step 1: Applying the LT to both sides of the equation and using linearity together with $\mathscr{L}\{2\}(s) = \frac{2}{s}$, we obtain

$$\mathscr{L}\{ty''\}(s) - \mathscr{L}\{ty'\}(s) + \mathscr{L}\{y\}(s) = \frac{2}{s}.$$

Step 2: Now using derivative of LT and LT of derivative theorems, we have

$$\mathscr{L}\{ty''\}(s) = -\frac{d}{ds}\mathscr{L}\{y''\}(s) = -\frac{d}{ds}(s^2Y(s) - sy(0) - y'(0)) = -2sY(s) - s^2Y'(s) + 2$$

and

$$\mathscr{L}\{ty'\}(s) = -\frac{d}{ds}\mathscr{L}\{y'\}(s) = -\frac{d}{ds}(sY(s) - y(0)) = -Y(s) - sY'(s),$$

where we use for simplicity $Y(s) = \mathscr{L}\{y\}(s)$.

Step 3: Substituting the latter in the equation and rearranging we find that

$$-2sY(s) - s^2Y'(s) + 2 + Y(s) + sY'(s) + Y(s) = \frac{2}{s}$$

or

$$Y'(s) + \frac{2}{s}Y(s) = \frac{2}{s^2}.$$

Step 4: Solving the linear 1st order differential equation above, we arrive at

$$Y(s) = \frac{2}{s} + \frac{C}{s^2},$$

where C is the constant of integration.

Step 5: Finally, inverse LT gives

$$y(t) = 2 + Ct.$$

Using $y'(0) = -1$, we see that $C = -1$ and so

$$y(t) = 2 - t.$$

Among the answers below choose the steps with mistake(s).

Select one or more:

- [x] Step 1 ✗
- [x] Step 2 ✗
- [x] Step 3 ✗
- [ ] Step 4
- [ ] Step 5
- [ ] there are no mistakes

Your answer is incorrect.

The correct answer is: there are no mistakes

Question 3

Incorrect

Mark 0.00 out of 1.00

Express the given function using step and window finctions:

$$g(t) = \begin{cases} 10, & 0 < t < 10 \\ 20, & 10 < t < 20 \\ 30, & t > 20 \end{cases}$$

Select one:

- ( ) $g(t) = 10u(t-10) + 20u(t-20) + 30\Pi_{20,30}(t)$
- ( ) $g(t) = 10u(t-10) + 20u(t-20) + 30u(t-20)$
- (•) $g(t) = 10\Pi_{0,10}(t) + 20\Pi_{10,20}(t) + 30\Pi_{20,30}(t)$ ✗
- ( ) $g(t) = 10\Pi_{0,10}(t) + 20\Pi_{10,20}(t) + 30u(t-20)$

Your answer is incorrect.

The correct answer is:

$$g(t) = 10\Pi_{0,10}(t) + 20\Pi_{10,20}(t) + 30u(t-20)$$

Question 4
Correct
Mark 1.00 out of 1.00

Find the Laplace transform of $\sin t \cdot u\left(t-\frac{\pi}{2}\right)$.

Select one:

- $e^{-\frac{\pi}{2}s} \cdot \frac{1}{s^2+1}$
- $e^{-\frac{\pi}{2}s} \cdot \frac{s}{s^2+1}$ ✓
- $-e^{-\frac{\pi}{2}s} \cdot \frac{1}{s^2+1}$
- $-e^{-\frac{\pi}{2}s} \cdot \frac{s}{s^2+1}$

Your answer is correct.

The correct answer is: $e^{-\frac{\pi}{2}s} \cdot \frac{s}{s^2+1}$

Question 5
Correct
Mark 1.00 out of 1.00

Find the inverse Laplace transform of $\frac{e^{-s}}{s^2+4}$.

Select one:

- $2u(t-1)\sin(t-1)$
- $\frac{1}{2}u(t-1)\sin(2t-2)$ ✓
- $\frac{1}{2}u(t-1)\sin(t-1)$
- $\frac{1}{2}u(t-1)\cos(2t-2)$

Your answer is correct.

The correct answer is: $\frac{1}{2}u(t-1)\sin(2t-2)$